

NFC batteryless e-ink tag

Recommended design: NT3H2211 (ISO 14443-A harvest + SRAM pass-through)\n+ STM32G071CBT6 + TPS61023 + GDEY042T81 4.2 in 400x300.\nPhone field powers the board. MCU streams the image, paints the panel, then the rail collapses.

Session: couple phone to the bezel antenna -> harvest charges VSTORE ->\nboost reaches 3.3 V -> supervisor releases PF2-NRST -> MCU boots ->\nSRAM mailbox drain over I2C -> enable EPD rail -> SSD1683 refresh -> deep sleep / field gone.

Do not hang bulk capacitance on NTAG VOUT. NXP limits that pin to 220 nF.\nCharge the tank through R1 (22 ohm) and a Schottky. Keep the phone on the coil for the whole refresh.

NFC front end



nfc.kicad_sch

Power



pwr.kicad_sch

MCU



mcu.kicad_sch

E-paper



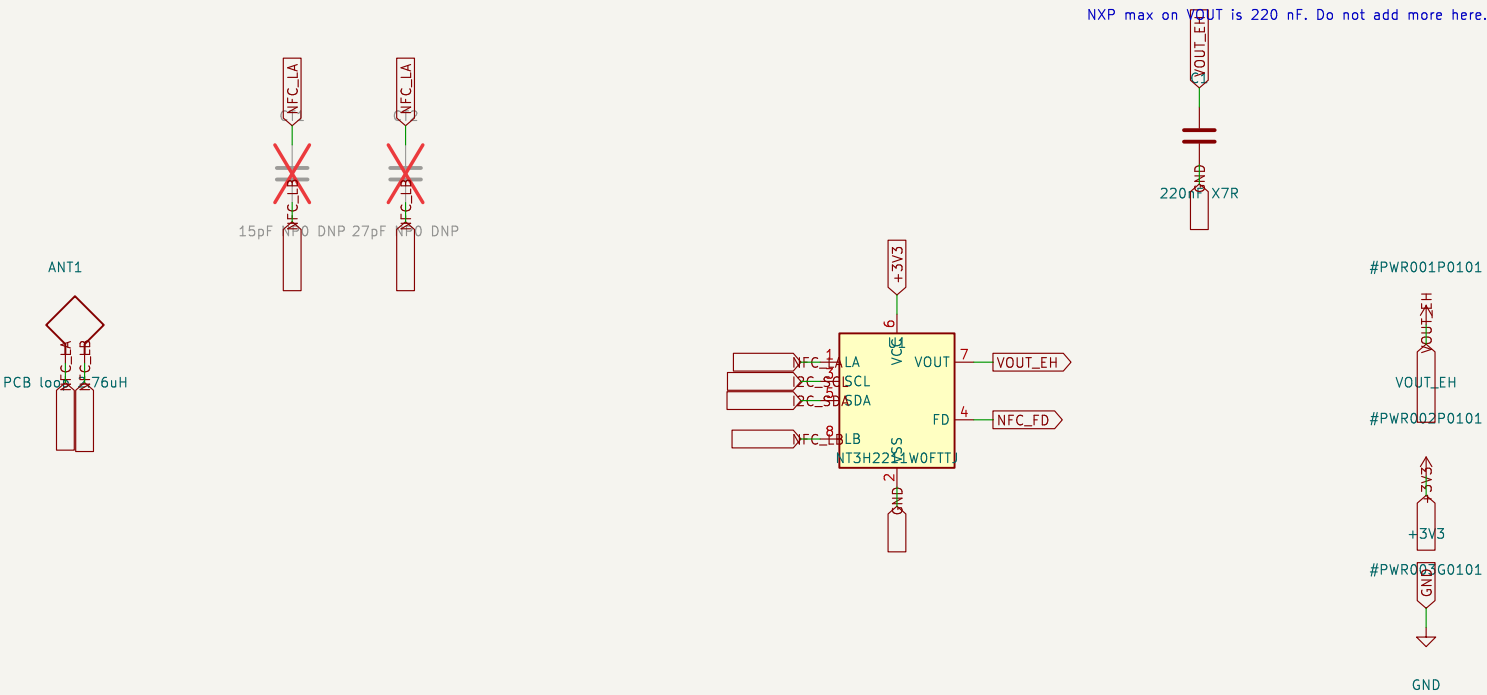
epd.kicad_sch

Board outline equals the GDEY042T81 module (91 mm x 77 mm).\nRoute a 2-turn Class-1 loop in the bezel; keep copper pour out of the antenna keepout.\nSee README.md for alternate designs (ST25DV, FM1280 module, 2.13 in, 5.83 in).

NFC batteryless e-ink tag		
playground		
Sheet: /		
File: nfc-eink.kicad_sch		
Title: NFC batteryless e-ink tag		
Size: A3	Date: 2026-09-04	Rev: 1
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 1/5	

Antenna and NT3H2211W0FTTJ

Internal Cr = 50 pF. Target Lant = 2.76 uH\nCT1/CT2 are NP0 tuning pads. Leave DNP until a VNA or phone-range sweep.



Harvest, tank, boost, EPD load, switch, reset

VOUT_EH -- D1 -- R1 -- VSTORE -- U2 boost --> +3V3

Hold off the EPD rail until VSTORE_DIV says the tank is ready.

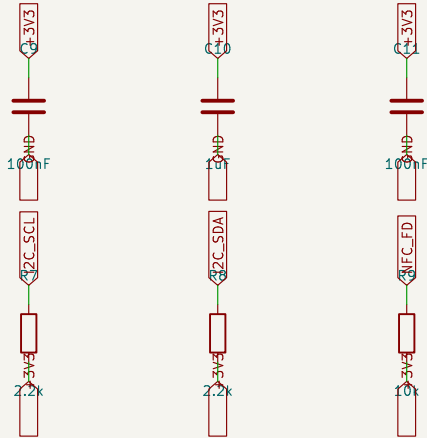
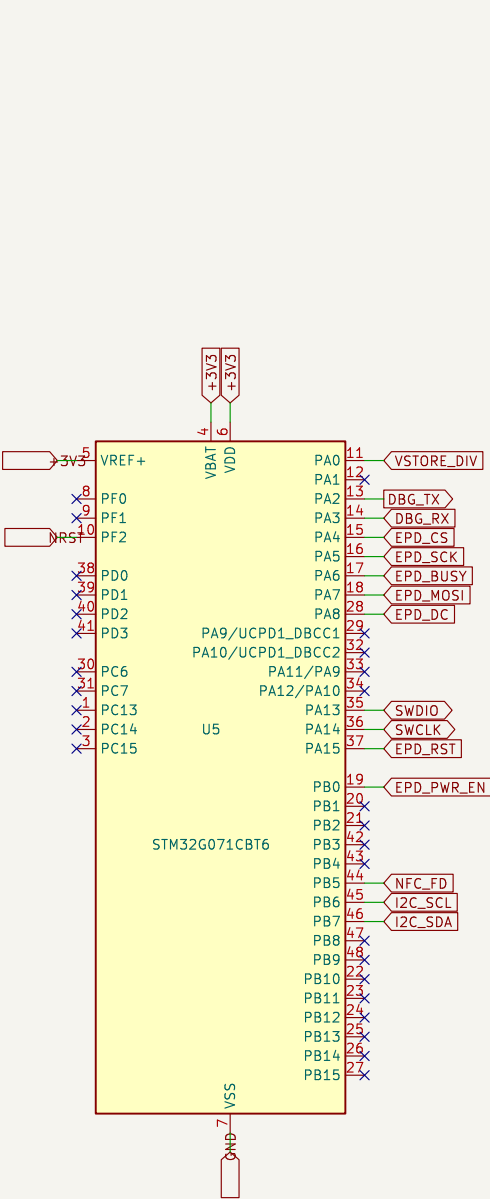
$V_{out} = 0.595 \times (1 + 453k/100k) = 3.29\text{ V}$

2.93 V threshold. Holds PF2–NRST until the boost is in regulation.

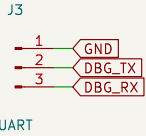
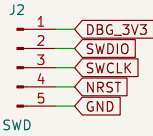
Close JP1 only when an ST–Link feeds J2 pin 1. Leave open for NFC power.

STM32G071CBT6 (HSI, no crystal)

36 KB RAM holds the 15 000-byte 4.2 in framebuffer.\nI2C1 on PB6/PB7. SPI1 on PA5/PA7. PF2 is nRESET on this package.

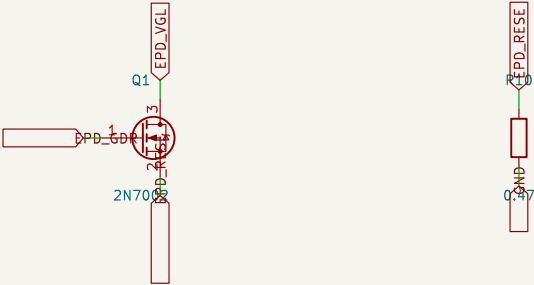
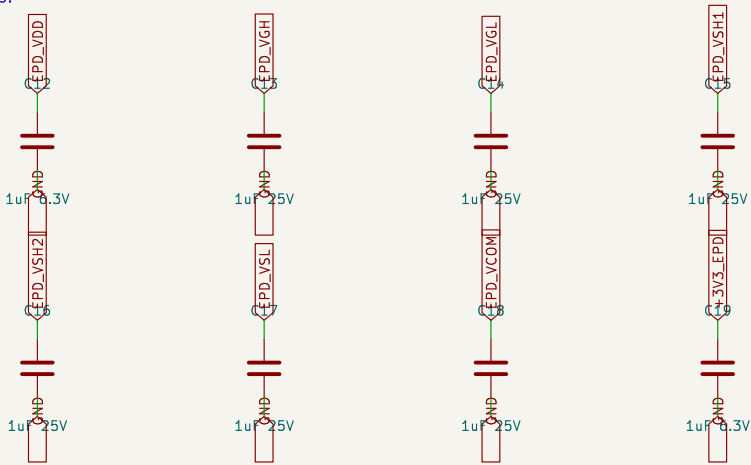


J2 1.27 mm: 3V3, SWDIO, SWCLK, NRST, GND



GDEY042T81 24-pin 0.5 mm FPC SSD1683

On-glass DC-DC still needs these host capacitors. Typical refresh is 13 mW / 5.6 mA at 3.0 V for 3 s.



SSD1683 booster sense MOSFET. Follow Good Display DESPI-C02 if a panel lot already has this on the FPC.